INCH-POUND
MIL-DTL-24135/12A(SH)
w/AMENDMENT 1
14 December 2021
SUPERSEDING
MIL-DTL-24135/12A(SH)
24 September 2020

DETAIL SPECIFICATION SHEET

HOSE, SYNTHETIC RUBBER, HIGH PERFORMANCE, WIRE REINFORCED (SINGLE-WIRE BRAID OVER SINGLE SYNTHETIC FIBER BRAID), FOR FLEXIBLE HOSE ASSEMBLIES (MEDIUM PRESSURE, HIGH TEMPERATURE, MULTI-FLUID, SIZES -4 THROUGH -40)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24135.

1. SCOPE

1.1 Scope. This specification covers high performance synthetic rubber hose, sizes -4 through -40, wire reinforced with a single-wire braid over a single layer of synthetic fiber braid that are used in flexible hose assemblies for medium pressure, high temperature, multi-fluid applications. This hose is compatible with petroleum, petroleum-based synthetic, and phosphate ester-based oils.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-2104	-	Lubricating Oil, Internal Combustion Engine, Combat/Tactical Service
MIL-PRF-17331	-	Lubricating Oil, Steam Turbine and Gear, Moderate Service
MIL-H-19457	-	Hydraulic Fluid, Fire-Resistant, Non-Neurotoxic
MIL-DTL-24135	-	Hose, Synthetic Rubber, Wire Reinforced, for Flexible Hose Assemblies, General Specifications for

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MIL-DTL-24787	-	Fittings, End, Reusable, for Flexible Hose Assemblies, General Specification for
MIL-DTL-24787/1	-	Fittings, End, Types F, FL, and FFL (Flange), Reusable, for Flexible Hose Assemblies
MIL-DTL-24787/2	-	Fittings, End, Type A (37-Degree Flare), Reusable, for Flexible Hose Assemblies
MIL-DTL-24787/3	-	Fittings, End, Types C and CL (O-Ring Seal Union), Reusable, for Flexible Hose Assemblies
MIL-DTL-24787/4	-	Fittings, End, Types E and EL (Split Clamp), Reusable, with Types SC (Split Clamp), TF (Tailpiece Female), and TM (Tailpiece Male) Adapters, for Flexible Hose Assemblies
MIL-DTL-24787/5	-	Fittings, End, Type D (Hose to Hose, 90-Degree Elbow), Reusable, for Flexible Hose Assemblies
MIL-DTL-32353	-	Hydraulic and Lubricating Oil, Synthetic Hydrocarbon Base
MIL-PRF-83282	-	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric, NATO Code Number H-537

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Hose furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Material. Rubber compounds used for hose shall be resistant to petroleum- and phosphate ester-based oils and seawater. The compounds shall consist of synthetic rubber, or a blend of synthetic rubbers, and shall be in accordance with MIL-DTL-24135 and as specified herein. Hose shall be colored blue.

3.3 Reinforcement. Wire reinforcement shall consist of a single-wire braid over a single layer of synthetic fiber braid applied in such a manner as to meet all the requirements of this specification.

3.4 Dimensions and pressures. Hose dimensions shall be in accordance with table I and hose pressures shall be in accordance with table II.

TABLE I. Hose dimensions.

Hose size (dash number)	Hose dimensions (inches)				Minimum bend radius (inches)
	Inside diameter (ID) tube		Outside diameter (OD) cover		
	Min	Max	Min	Max	
-4	0.188	0.214	0.500	0.539	3.00
-5	0.250	0.281	0.570	0.601	3.38
-6	0.312	0.343	0.656	0.695	4.00
-8	0.406	0.437	0.743	0.789	4.62

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TABLE I. Hose dimensions – Continued.

-10	0.500	0.539	0.899	0.945	5.50
-12	0.625	0.667	1.055	1.101	6.50
-16	0.875	0.917	1.203	1.266	7.38
-20	1.125	1.172	1.469	1.531	9.00
-24	1.375	1.422	1.719	1.781	10.50
-32	1.812	1.859	2.190	2.266	13.25
-40	2.375	2.437	2.828	2.922	24.00

TABLE II. Hose pressures and weight.

Hose size (dash number)	Pressure (psi)		Impulse peak pressure (psi)	Approximate weight/feet (lbs/ft)	Vacuum (inches of Hg)
	Working (max)	Burst (min)			
-4	3,000	12,000	3,750	0.15	28
-5	3,000	12,000	3,750	0.19	28
-6	2,250	9,000	2,813	0.23	28
-8	2,000	8,000	2,500	0.28	28
-10	1,750	7,000	2,188	0.39	28
-12	1,500	6,000	1,875	0.47	28
-16	800	3,200	1,000	0.41	20
-20	625	2,500	625	0.59	20
-24	500	2,000	500	0.72	15
-32	300	1,200	300	0.90	11
-40	300	1,200	300	1.63	11

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-DTL-24135 and 4.4.

4.3 Conformance inspection. Conformance inspection shall be in accordance with MIL-DTL-24135.

4.4 Sampling for qualification testing.

4.4.1 Hose. Hose samples for qualification testing shall be as specified in table III. Samples shall be not less than 15 inches in length and shall be fitted with end fittings qualified in accordance with MIL-DTL-24787.

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TABLE III. Hose and end fitting samples required.

Hose size (dash number)	Number of hose samples to be provided	Number of end fitting samples to be provided 1/
-8	6	12
-32 or -40	7 2/	14

NOTES:
1/ End fittings for testing shall be qualified in accordance with MIL-DTL-24787.
2/ Ensure there is an adequate amount of hose to conduct flame spread testing (see 4.5.11 of MIL-DTL-24135).

4.4.2 End fittings. End fittings used for the qualification of hose to this specification sheet shall be end fittings qualified in accordance with MIL-DTL-24787, using any of the end fitting types identified in table IV. The quantity of end fittings required shall be in accordance with table III. The end fitting supplier and hose supplier may be a single source. If a combined end fitting and hose qualification is pursued, end fittings in accordance with MIL-DTL-24787/2 or /3 shall be used for the hose qualification portion to assist with end fitting qualification. Upon receiving qualification of this hose, the hose shall be considered qualified for use with all end fitting types identified in table IV.

TABLE IV. MIL-DTL-24787 end fittings for use with MIL-DTL-24135/12 hoses.

End fitting specification sheet number	End fitting type per specification sheet	End fitting description
24787/1 1/	F, FL, FFL	Flange to hose (straight, 90-degree elbow, or 45-degree elbow)
24787/2	A	37-degree flare swivel
24787/3 2/	C, CL	O-ring seal union (straight or 90-degree elbow)
24787/4 3/	E, EL, SC, TF, TM	Split-clamp (straight, 90-degree elbow, or assembly). Tailpiece (female or male)
24787/5 2/	D	90-degree elbow

NOTES:
1/ For -12 to -32 hose dash sizes only.
2/ Does not include -4 hose dash sizes.
3/ For -32 and -40 hose dash sizes only.

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4.4.2.1 Hose and end fitting mix. Where there are two or more qualified suppliers of end fittings in accordance with MIL-DTL-24787, or when the prospective hose supplier is also qualifying an end fitting, the hose and end fitting mix shall be in accordance with 4.4.2.1.1 through 4.4.2.1.3 below, as appropriate. When qualifying a hose and end fitting, hose samples shall be coupled on one end with the end fitting being submitted for qualification and on the other end with a previously qualified end fitting.

4.4.2.1.1 Qualifying hose with multiple qualified end fitting suppliers. Where there are two or more qualified suppliers of end fittings, the end fitting mix shall be in accordance with table V.

TABLE V. End fitting mix when there are multiple qualified end fitting suppliers for hose qualification test.

Sample size 1/ (dash number)	Number of fittings to be provided 2/	
	Qualified source #1	Qualified source #2
-8	8	4
-32 or -40	10	4

NOTES:

1/ Satisfactory completion of qualification testing of the samples selected in accordance with tables I, II, and III will qualify the manufacturer for all sizes in this specification for the end fitting specification sheet(s) in table IV.

2/ See table IV for end fitting options.

4.4.2.1.2 Qualifying end fitting and hose with one qualified end fitting supplier and one qualified hose supplier. Where the prospective hose supplier is also qualifying an end fitting, and there is only one qualified supplier of end fittings, hose, or both, the end fitting and hose mix shall be in accordance with the section of the same title in MIL-DTL-24787/2 or /3. For end fittings being qualified at the same time, the sequence of tests for end fitting qualification shall be in accordance with MIL-DTL-24787/2 or /3. The sequence of tests for hose qualification shall be in accordance with 4.5. The sample size of hose dash of -8 represents the smallest sample size in MIL-DTL-24787/2 or /3, while hose dash of -40 represents the largest sample size.

4.4.2.1.3 Qualifying end fitting and hose with two or more qualified end fitting and qualified hose suppliers. Where the prospective hose supplier is also qualifying an end fitting, and there are two or more qualified suppliers of end fittings, hose, or both, the end fitting and hose mix shall be in accordance with the section of the same title in MIL-DTL-24787/2 or /3. For end fittings being qualified at the same time, the sequence of tests for end fitting qualification shall be in accordance with MIL-DTL-24787/2 or /3. The sequence of tests for hose qualification shall be in accordance with 4.5. The sample size of hose dash of -8 represents the smallest sample size in accordance with MIL-DTL-24787/2 or /3, while hose dash of -40 represents the largest sample size.

4.5 Test sequence. The sequence of tests on sample hose assemblies shall be as specified in table VI.

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TABLE VI. Sequence of qualification tests on sample assemblies.

Test sequence	Sample numbers												
	Size -8						Size -32 or -40						
	1	2	3	4	5	6	7	8	9	10	11	12	13
Proof pressure (see 4.5.1 1/)	X	X	X	X	X	--	X	X	X	X	X	--	--
Stability (see 4.5.2 1/)	X	X	X	X	X	--	X	X	X	X	X	--	--
Aging (see 4.5.3 1/)	X	X	X	X 2/	--	--	X	X	X	X 2/	--	--	--
Oil circulation (see 4.7 3/)	--	X	--	--	--	--	--	X	--	--	--	--	--
Impulse (see 4.5.5 1/)	--	--	--	--	X	--	--		--	--	X	--	--
Burst pressure (see 4.5.6 1/)	X	--	--	--	--	--	X	--	--	--	--	--	--
Vacuum (see 4.9 3/)	--	--	--	X 2/	--	--	--	--	--	X 2/	--	--	--
Cold temperature flexibility (see 4.5.8 1/)	--	--	X	--	--	--	--	--	X	--	--	--	--
Adhesion (see 4.5.9 1/)	--	--	--	X		--	--	--	--	X	--	--	--
Fire resistance (see 4.5.10 1/)	--	--	--	--	--	X	--	--	--	--	--	X	X

NOTES:
1/ See specified section in MIL-DTL-24135 for test details.
2/ See 4.9 for details on conducting test prior to and following aging of this sample hose.
3/ See specified section within this specification sheet.

4.6 Test fluid. The test fluid shall be oil or hydraulic fluid in accordance with MIL-PRF-2104, MIL-PRF-17331, MIL-DTL-32353, MIL-PRF-83282, or MIL-H-19457 for the aging, oil circulation, impulse, and cold temperature tests.

4.7 Oil circulation. Sample hose assemblies shall be filled with oil fluid in accordance with 4.6. Assemblies shall be installed in a 180-degree arc and tested on a flex and circulation test stand as shown on figure 1. The radius at the inside of the bend shall be equal to the applicable arc and flex bend radius specified in table I. Oil maintained at 300±5 °F shall be circulated through the assemblies at 60 psi, at a flow rate of not less than 3 gallons per minute, while the assemblies are flexed 3.5 inches in each direction at 12±1 cycles per minute (CPM). The hose assembly shall be subjected to 400 hours of hot oil circulation and flexing without evidence of leakage or other malfunction. The oil shall be changed after 200 hours of circulation.

4.8 Impulse pressure. The peak pressure for the impulse test shall be 125 percent of the maximum working pressure in accordance with table II for sizes -4 through -16 and 100 percent of the maximum working pressure for sizes -20 through -40. The pressure time curves shall be in accordance with figure 1 or 2, as applicable. Impulse testing shall be conducted at 300 °F and shall be for one million cycles. Sizes up to -16 shall be bent to 90 degrees and sizes -20 and larger shall be straight with one end free. The radius at the inside of the arc of sizes up to -16 shall be equal to the minimum bend specified in table I. The rate of impulsing shall be 70±35 CPM with the pressure time curve in accordance with figure 1 or 2, as applicable. The test fluid shall be in accordance with 4.6. Heating or cooling of the test chamber is not permitted to maintain the required test temperature. For each size sample tested, failure to meet one of the following conditions shall constitute failure for that group:

a. Failure of any one test assembly to attain 750,000 impulse cycles.
b. Failure of any group to average one million impulse cycles.
   (1) No average greater than 1,250,000 cycles shall be included.
   (2) The cycles from no greater than six assemblies shall be included in the calculation.

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4.9 Vacuum. Vacuum testing shall be performed both prior to and after oil aging. Sample sizes -4 through -16 shall be bent to 180 degrees and sizes -20 through -40 to 90 degrees to the minimum bend radius specified in table I and the OD remeasured at the flattened section. Deformation of the OD in excess of 10 percent of the original OD shall be cause for rejection of the sample. With the hose samples held in the bent position, a vacuum as specified in table II shall be applied and held for 5 minutes. The hose shall be rechecked for additional flattening and for leakage as evidence by loss of vacuum. Application of the vacuum shall not result in greater than an additional 10 percent reduction in the flattened section of the OD for sizes up to and including -24, and 25 percent reduction for sizes -32 and larger. Upon completion of the vacuum test, the sample assemblies shall be subjected to a 350-psi pressure test with no leakage from the hose or fittings permitted. The sample hoses shall then be dissected and examined. Any evidence of ply separation, blistering, collapse, or other damage shall be cause for rejection.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Hoses are intended for use in piping systems as flexible connections to resiliently mounted (sound mounted) equipment that may be used for other applications, which do not exceed the pressure or temperature limits of the hose selected, and have an established service life as documented in S6430-AE-TED-010, Piping Devices, Flexible Hose Assemblies. Hose and hose assemblies procured to this specification are form, fit, and function replacements for prior revisions. Existing stock of hose and hose assemblies manufactured to MIL-H-24135B, dated 3 September 1993, are acceptable for use until depleted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Size of hose (see 3.4).

c. Quantity of hose (in linear feet) per size.

d. Type and size of end fitting for hose assembly (see table IV and MIL-DTL-24787).

e. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 24135 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Subject term (key word) listing.

37-degree flare swivel to hose, straight

90-degree elbow, hose to hose

Flange to hose, 45-degree elbow

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Flange to hose, 90-degree elbow

Flange to hose, straight

O-ring seal union, male half to hose, 90-degree elbow

O-ring seal union, male half to hose, straight

Petroleum

Phosphate esters

Seawater

Split clamp assembly

Split clamp to hose, 90-degree elbow

Split clamp to hose, straight

Synthetic oil

Tailpiece, female, split clamp, silver braze

Tailpiece, male, split clamp, silver braze

6.5 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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HOSE CUT LENGTH (HCL) = R+2J+3.5+20D

$R = r + \frac{OD}{2}$

r = MINIMUM BEND RADIUS

J = SOCKETED HOSE

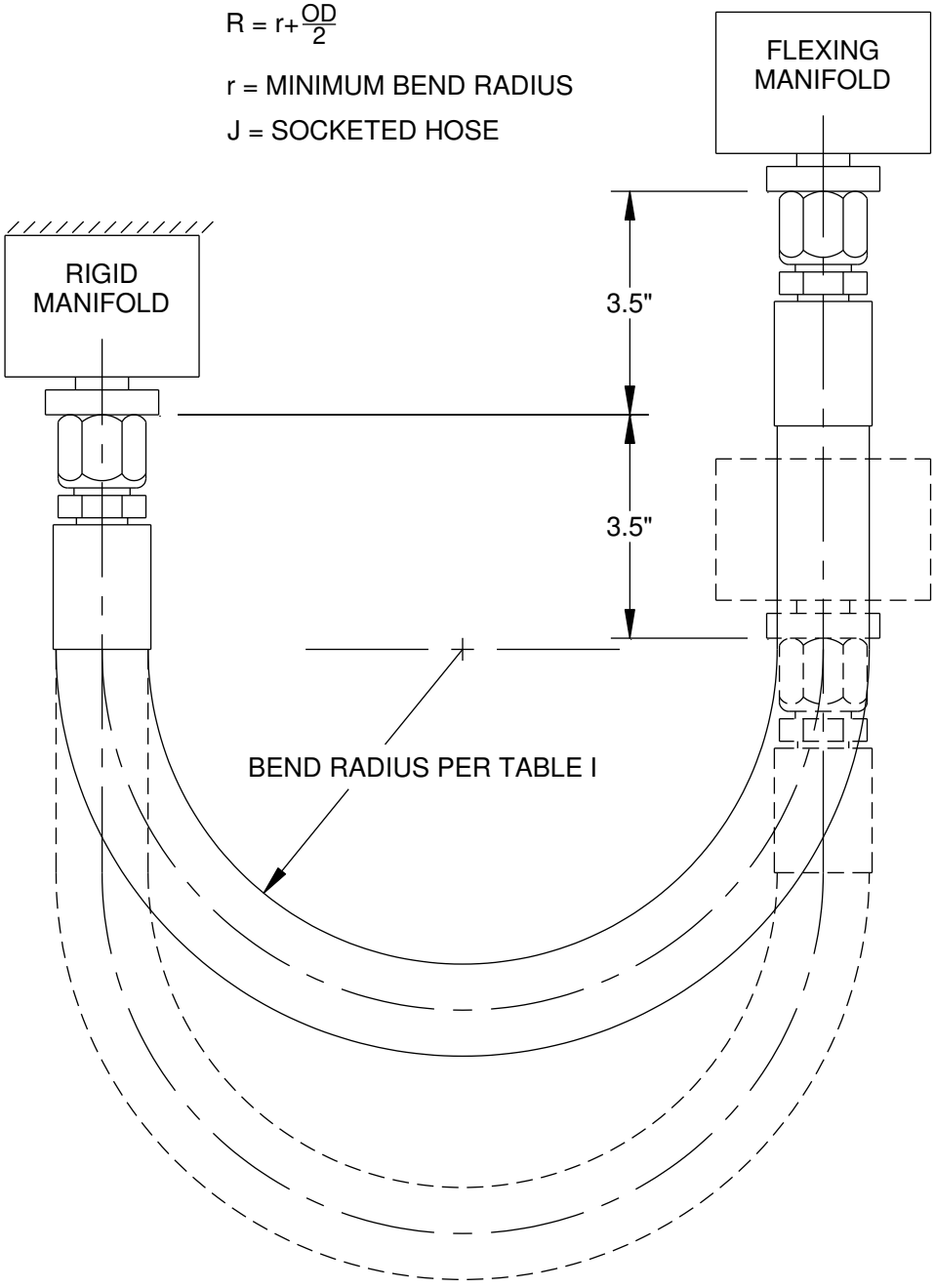


FIGURE 1. Flex and circulation test stand.

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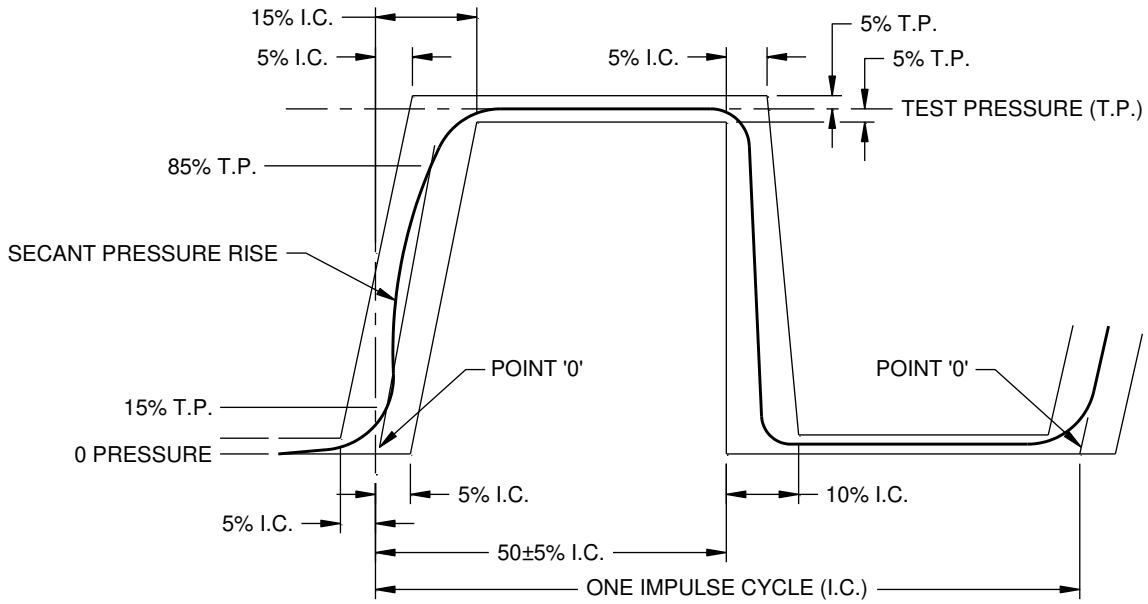


FIGURE 2. Impulse pressure curve.

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CONCLUDING MATERIAL

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